

15 *Basic Acceptance Sampling Procedures*

CHAPTER OUTLINE

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|---|--|
| 15.1 THE ACCEPTANCE-SAMPLING PROBLEM | 15.3.2 Multiple-Sampling Plans |
| 15.1.1 Advantages and Disadvantages of Sampling | 15.3.3 Sequential-Sampling Plans |
| 15.1.2 Types of Sampling Plans | 15.4 MILITARY STANDARD 105E (ANSI/ASQC Z1.4, ISO 2859) |
| 15.1.3 Lot Formation | 15.4.1 Description of the Standard |
| 15.1.4 Random Sampling | 15.4.2 Procedure |
| 15.1.5 Guidelines for Using Acceptance Sampling | 15.4.3 Discussion |
| 15.2 SINGLE-SAMPLING PLANS FOR ATTRIBUTES | 15.5 THE DODGE-ROMIG SAMPLING PLANS |
| 15.2.1 Definition of a Single-Sampling Plan | 15.5.1 AOQL Plans |
| 15.2.2 The OC Curve | 15.5.2 LTPD Plans |
| 15.2.3 Designing a Single-Sampling Plan with a Specified OC Curve | 15.5.3 Estimation of Process Average |
| 15.2.4 Rectifying Inspection | Supplemental Material for Chapter 15 |
| 15.3 DOUBLE, MULTIPLE, AND SEQUENTIAL SAMPLING | S15.1 A Lot Sensitive Compliance (LTPD) Sampling Plan |
| 15.3.1 Double-Sampling Plans | S15.2 Consideration of Inspection Error |

Learning Objectives

1. Understand the role of acceptance sampling in modern quality control systems
2. Understand the advantages and disadvantages of sampling
3. Understand the difference between attributes and variables sampling plans, and the major types of acceptance-sampling procedures
4. Know how single-, double-, and sequential-sampling plans are used
5. Understand the importance of random sampling
6. Know how to determine the OC curve for a single-sampling plan for attributes
7. Understand the effects of the sampling plan parameters on sampling plan performance
8. Know how to design single-sampling, double-sampling, and sequential-sampling plans for attributes
9. Know how rectifying inspection is used
10. Understand the structure and use of MIL STD 105E and its civilian counterpart plans
11. Understand the structure and use of the Dodge–Romig system of sampling plans

15.1 The Acceptance Sampling Problem

- Typical application of acceptance sampling is for **lot disposition** (accept or reject the lot), sometimes referred to as **lot sentencing**, for receiving inspection activities.
 - Accepted lots are put into production
 - Rejected lots may be returned to supplier or subjected to other **lot-disposition action**
- Sampling methods may also be used during various stages of production.
- Three aspects of sampling are important:
 1. It is the purpose of acceptance sampling **to sentence lots, not to estimate the lot quality.**
 2. Acceptance sampling **simply accepts and rejects lots is not used to control or improve quality.**
 3. It is **an audit tool to ensure that the output of a process conforms to requirements.**

15.1 The Acceptance Sampling Problem

- There are three approaches to lot sentencing:
 1. accept with no inspection;
 2. 100% inspection—that is, inspect every item in the lot, removing all defective1 units found (defectives may be returned to the supplier, reworked, replaced with known good items, or discarded); and
 3. acceptance sampling.
- Acceptance sampling is most likely to be useful in the following situations:
 1. When testing is destructive
 2. When the cost of 100% inspection is extremely high
 3. When 100% inspection is not technologically feasible (require so much time)
 4. When there are many items to be inspected and the inspection error rate is high
 5. When the supplier has an excellent quality history, but the supplier's process capability is sufficiently low
 6. When there are potentially serious product liability risks

15.1.1 Advantages and Disadvantages of Sampling

When acceptance sampling is contrasted with 100% inspection, it has the following advantages:

1. It is usually less expensive because there is less inspection.
2. There is less handling of the product, hence reduced damage.
3. It is applicable to destructive testing.
4. Fewer personnel are involved in inspection activities.
5. It often greatly reduces the amount of inspection error.
6. The rejection of entire lots as opposed to the simple return of defectives often provides a stronger motivation to the supplier for quality improvements.

Acceptance sampling also has several disadvantages, however. These include the following:

1. There are risks of accepting “bad” lots and rejecting “good” lots.
2. Less information is usually generated about the product or about the process that manufactured the product.
3. Acceptance sampling requires planning and documentation of the acceptance-sampling procedure whereas 100% inspection does not.

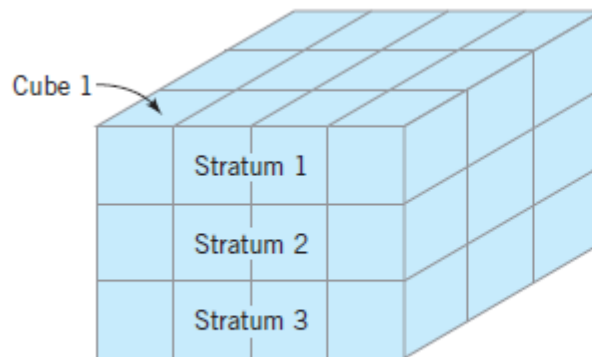
Types of sampling plans

- One major classification is by data type, **variables** and **attributes**
- Another is based on the number of samples required for a decision. These include:
 - Single-sampling plans (accept or reject)
 - Double-sampling plans (accept, reject or take a second sample one more time to judge)
 - Multiple-sampling plans (accept, reject or take a multiple samples to judge)
 - Sequential-sampling plans (take samples until you accept or reject)
- Single-, double-, multiple-, and sequential sampling plans can be designed to produce equivalent results.
- Factors to consider when selecting the type of sampling procedure include:
 - Administrative efficiency
 - Type of information produced by the plan
 - Average amount of inspection required by plan
 - Impact of the procedure on manufacturing flow

Lot Formation and Random Sampling

- There are a number of important considerations informing lots for inspection, including:
 1. Lots should be **homogeneous**.
 2. **Larger lots are preferred over smaller ones.**
 3. Lots should be **conformable to materials-handling** systems used in both supplier and consumer facilities.
- The units selected for inspection from the lot **should be chosen at random**, and they should be representative of all the items in the lot. **The random-sampling concept is extremely important in acceptance sampling.**
- Sometimes the inspector may **stratify** the lot. This consists of dividing the lot into strata or layers and then subdividing each strata into cubes, as shown in the Figure

Units are then selected from within each cube.



- Although this stratification of the lot is usually an **imaginary activity** performed by the inspector but at least it ensures that units are selected from all locations in the lot.

■ FIGURE 15.1 Stratifying a lot.

15.1.5 Guidelines for Using Acceptance Sampling

- An **acceptance-sampling plan** consists of sample size and acceptance/rejection criteria for lot sentencing
- An **acceptance-sampling scheme** is a set of procedures consisting of acceptance-sampling plans in which lot sizes, sample sizes, and acceptance/rejection criteria along with amount of 100% inspection and sampling are related

- A **sampling scheme** is a set of acceptance-sampling plans in which lot sizes, sample sizes, and acceptance/rejection criteria along with amount of 100% inspection and sampling are related.
 - **AQL** Acceptable Quality Level A “good” quality level considered satisfactory. LTPD Lot Tolerance Percent Defective represents a poor-quality level the consumer wants protection against.
 - **AOQL** Average Outgoing Quality Limit Maximum average defect level after inspection.

Objective	Attributes Procedure	Variables Procedure
Ensure quality levels for consumer/producer	Select plan for specific OC curve	Select plan for specific OC curve
Maintain quality at a target	AQL system; MIL STD 105E, ANSI/ASQC Z1.4	AQL system; MIL STD 414, ANSI/ASQC Z1.9
Ensure average outgoing quality level	AOQL system; Dodge–Romig plans	AOQL system
Reduce inspection, with small sample sizes, good-quality history	Chain sampling	Narrow-limit gauging
Reduce inspection after good-quality history	Skip-lot sampling; double sampling	Skip-lot sampling; double sampling
Ensure quality no worse than target	LTPD plan; Dodge–Romig plans	LTPD plan; hypothesis testing

15.1.5 Guidelines for Using Acceptance Sampling

■ **TABLE 15.1**

Acceptance-Sampling Procedures

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AQL Acceptable Quality Level A “good” quality level considered satisfactory.

LTPD Lot Tolerance Percent Defective represents a poor-quality level the consumer wants protection against.

AOQL Average Outgoing Quality Limit Maximum average defect level after inspection.

15.2 Single-Sampling Plans for Attributes

15.2.1 Definition of a Single-Sampling Plan

Suppose that a lot of size N has been submitted for inspection. A **single-sampling plan** is defined by the sample size n and the acceptance number c . Thus, if the lot size is $N = 10,000$, then the sampling plan

$$n = 89$$

$$c = 2$$

means that from a lot of size 10,000 a random sample of $n = 89$ units is inspected and the number of nonconforming or defective items d observed. If the number of observed defectives d is less than or equal to $c = 2$, the lot will be accepted. If the number of observed defectives d is greater than 2, the lot will be rejected. Since the quality characteristic inspected is an attribute, each unit in the sample is judged to be either conforming or nonconforming. One or several attributes can be inspected in the same sample; generally, a unit that is nonconforming to specifications on one or more attributes is said to be a defective unit. This procedure is called a single-sampling plan because the lot is sentenced based on the information contained in one sample of size n .

15.2.2 The OC Curve

An important measure of the performance of an acceptance-sampling plan is the **operating-characteristic (OC) curve**. This curve plots the probability of accepting the lot versus the lot fraction defective. Thus, the OC curve displays the discriminatory power of the sampling plan; that is, it shows the probability that a lot submitted with a certain fraction defective will be either accepted or rejected. The OC curve of the sampling plan $n = 89$, $c = 2$ is shown in Figure 15.2. It is easy to demonstrate how the points on this curve are obtained. Suppose that the lot size N is large (theoretically infinite). Under this condition, the distribution of the number of defectives d in a random sample of n items is binomial with parameters n and p , where p is the fraction of defective items in the lot. An equivalent way to conceptualize this is to draw lots of N items at random from a theoretically infinite process, and then to draw random samples of n from these lots. Sampling from the lot in this manner is the equivalent of sampling directly from the process. The probability of observing exactly d defectives is

$$P\{d \text{ defectives}\} = f(d) = \frac{n!}{d!(n-d)!} p^d (1-p)^{n-d} \quad (15.1)$$

The probability of acceptance is simply the probability that d is less than or equal to c , or

$$P_a = P\{d \leq c\} = \sum_{d=0}^c \frac{n!}{d!(n-d)!} p^d (1-p)^{n-d} \quad (15.2)$$

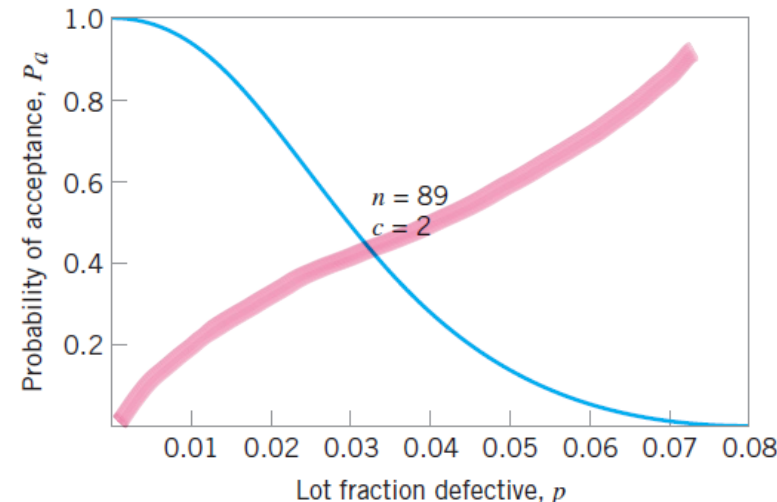
For example, if the lot fraction defective is $p = 0.01$, $n = 89$, and $c = 2$, then

$$\begin{aligned} P_a &= P\{d \leq 2\} = \sum_{d=0}^2 \frac{89!}{d!(89-d)!} (0.01)^d (0.99)^{89-d} \\ &= \frac{89!}{0!89!} (0.01)^0 (0.99)^{89} + \frac{89!}{1!88!} (0.01)^1 (0.99)^{88} + \frac{89!}{2!(87)!} (0.01)^2 (0.99)^{87} \\ &= 0.9397 \end{aligned}$$

The OC curve is developed by evaluating equation 15.2 for various values of p . Table 15.2 displays the calculated value of several points on the curve.

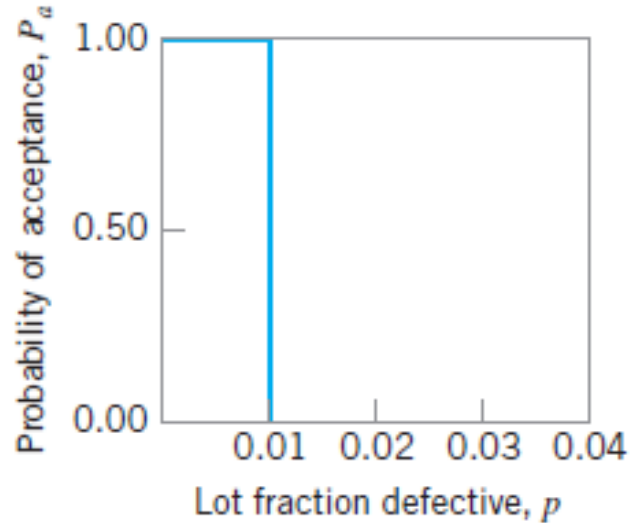
The OC curve shows the **discriminatory power** of the sampling plan. For example, in the sampling plan $n = 89$, $c = 2$, if the lots are 2% defective, the probability of acceptance is approximately 0.74. This means that if 100 lots from a process that manufactures 2% defective product are submitted to this sampling plan, we will expect to accept 74 of the lots and reject 26 of them.

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ممکن ہے رسم خطاً و نفس لیه

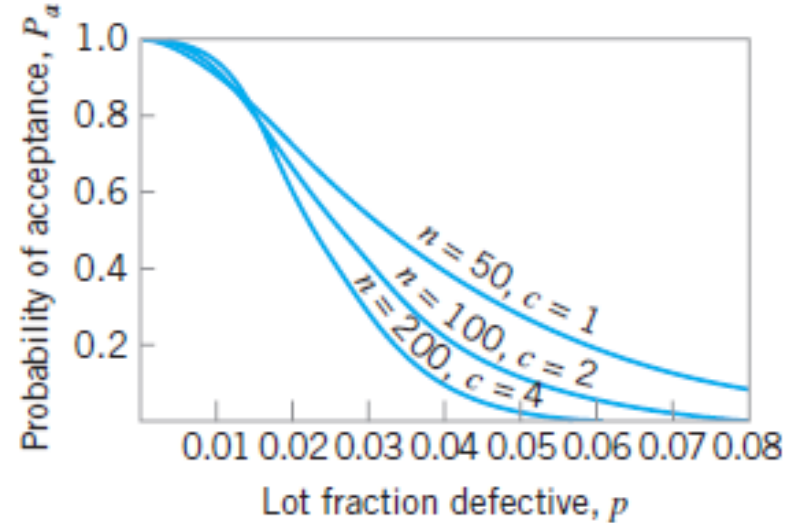


■ **FIGURE 15.2** OC curve of the single-sampling plan $n = 89$, $c = 2$.

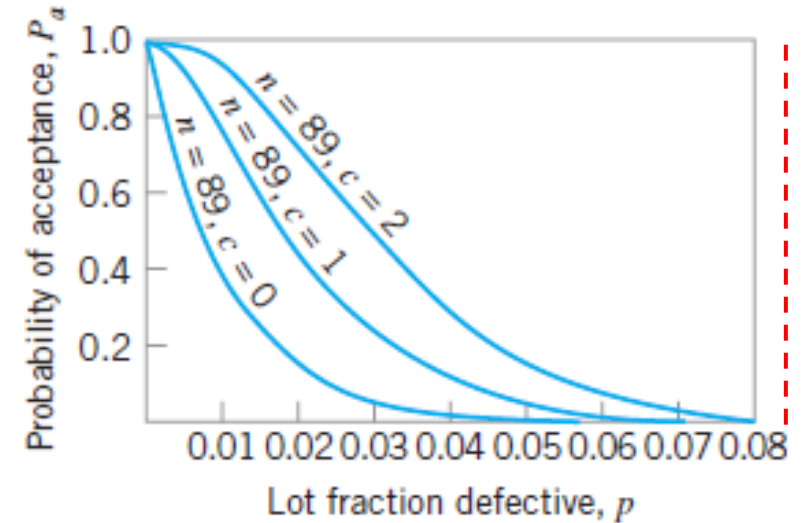
Effect of n and c on OC curves:



■ **FIGURE 15.3** Ideal OC curve.



■ **FIGURE 15.4** OC curves for different sample sizes.



■ **FIGURE 15.5** The effect of changing the acceptance number on the OC curve.

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Specific Points on the OC Curve

- The poorest quality level for the supplier's process that a consumer would consider to be acceptable as a process average is called the **acceptable quality level (AQL)**
- **AQL is a property of the supplier's manufacturing process, not a property of the sampling plan**
- The protection obtained for individual lots of poor quality is established by the **lot tolerance percent defective (LTPD)**
- Also called **rejectable quality level (RQL)** and the **limiting quality level (LQL)**
- **LTPD is a level of lot quality specified by the consumer**, not a characteristic of the sampling plan
- Sampling plans can be designed to have specified performance at the AQL and the LTPD points

15.2.3 Designing a Single-Sampling Plan with a Specified OC Curve

A common approach to the design of an acceptance-sampling plan is to require that the OC curve pass through two designated points. Note that one point is not enough to fully specify the sampling plan; however, two points are sufficient. In general, it does not matter which two points are specified.

Suppose that we wish to construct a sampling plan such that the probability of acceptance is $1 - \alpha$ for lots with fraction defective p_1 , and the probability of acceptance is β for lots with fraction defective p_2 . Assuming that binomial sampling (with type-B OC curves) is appropriate, we see that the sample size n and acceptance number c are the solution to

Type I error = P(reject good lot)

Type II error = P(accept bad lot)

$$\left. \begin{aligned} 1 - \alpha &= \sum_{d=0}^c \frac{n!}{d!(n-d)!} p_1^d (1-p_1)^{n-d} \\ \beta &= \sum_{d=0}^c \frac{n!}{d!(n-d)!} p_2^d (1-p_2)^{n-d} \end{aligned} \right\} \quad (15.3)$$

Designing a Single-Sampling Plan with a Specified OC Curve

- Use a chart called a **Binomial Nomograph** to design plan

- Specify:

- p_1 = AQL (Acceptable Quality Level)

- p_2 = LTPD (Lot Tolerance Percent Defective)

- $1 - \alpha$ = P[Lot is accepted | p = AQL] → ممكن ما تعطيني الا حتى لات من الرسم طالع OC
لل acceptance و الـ curve واقول اذا هو OC أولا

- β = P[Lot is accepted | p = LTPD]

مثال سلايد ١٢

Use a Binomial Nomograph to Find Sampling Plan

(Figure 15-9, p. 643)

20

- Draw two lines on nomograph
 - Line 1 connects $p_1 = \text{AQL}$ to $(1 - \alpha)$
 - Line 2 connects $p_2 = \text{LTPD}$ to β
 - Pick n and c from the intersection of the lines
- Example: Suppose
 - $p_1 = 0.01$,
 - $\alpha = 0.05$, $\rightarrow 1 - \alpha = 1 - 0.05 = 0.95$
 - $p_2 = 0.06$,
 - $\beta = 0.10$.

Find the acceptance sampling plan.

$p_1 = AQL = .01$

$p_2 = LTPD = .06$

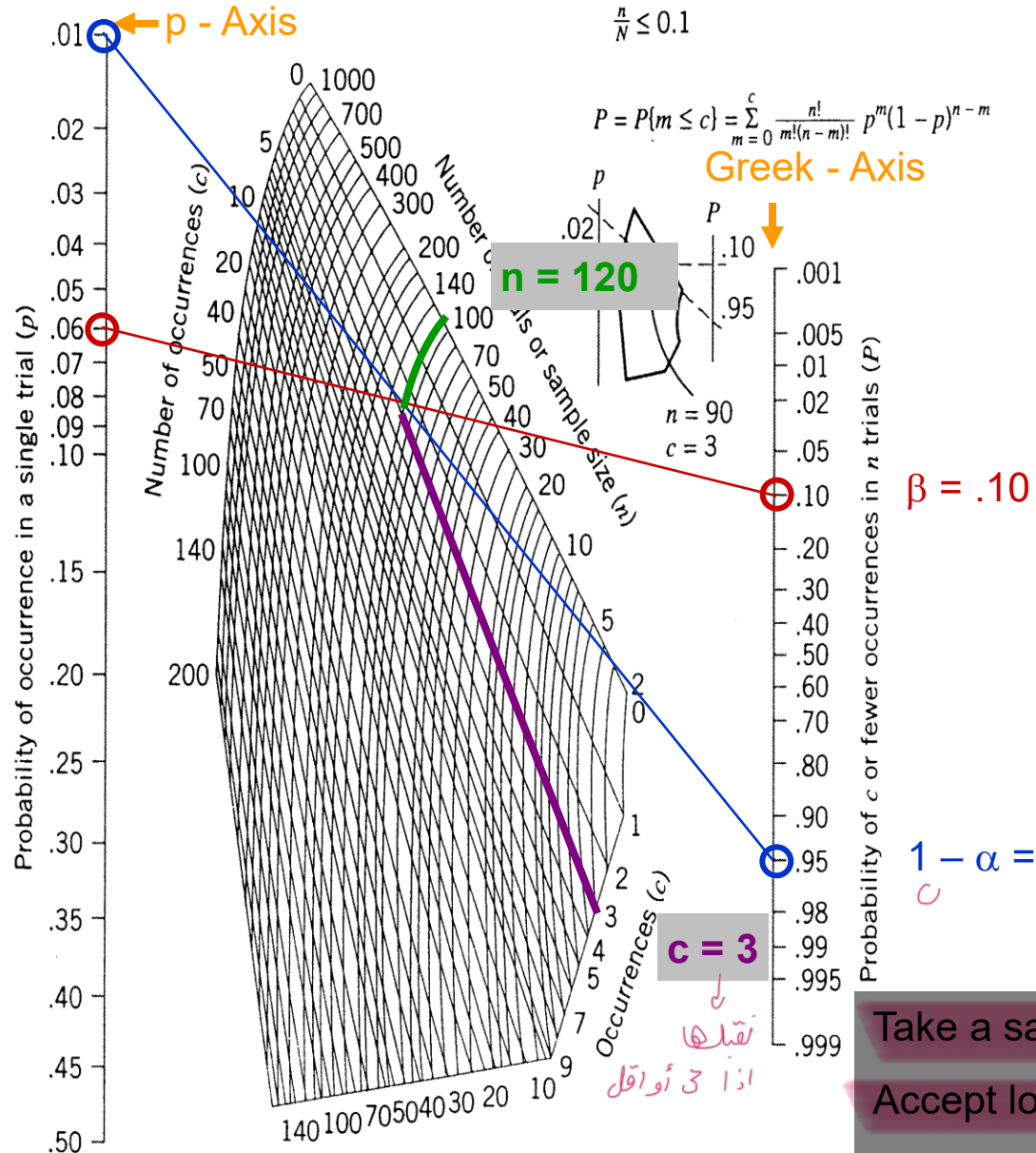
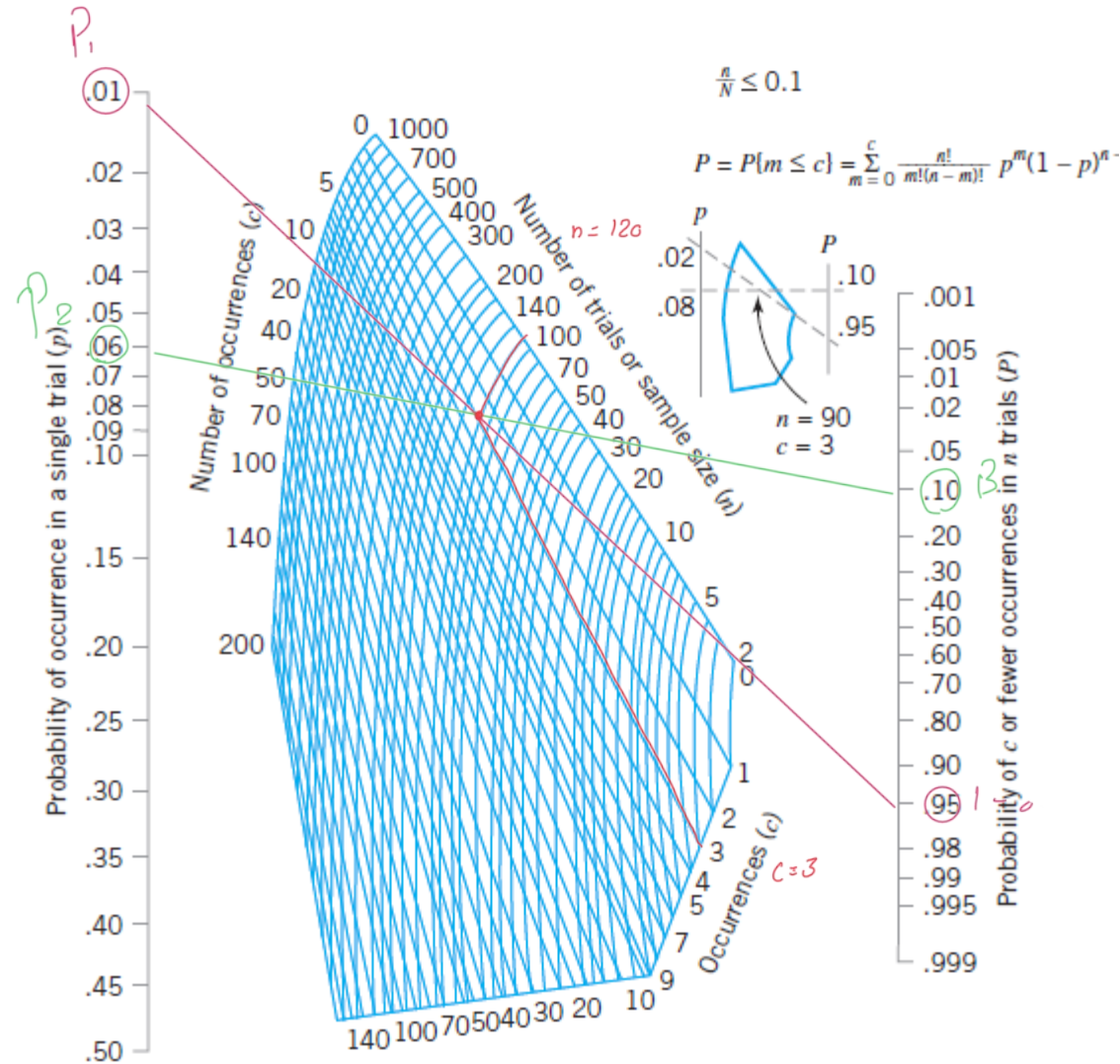


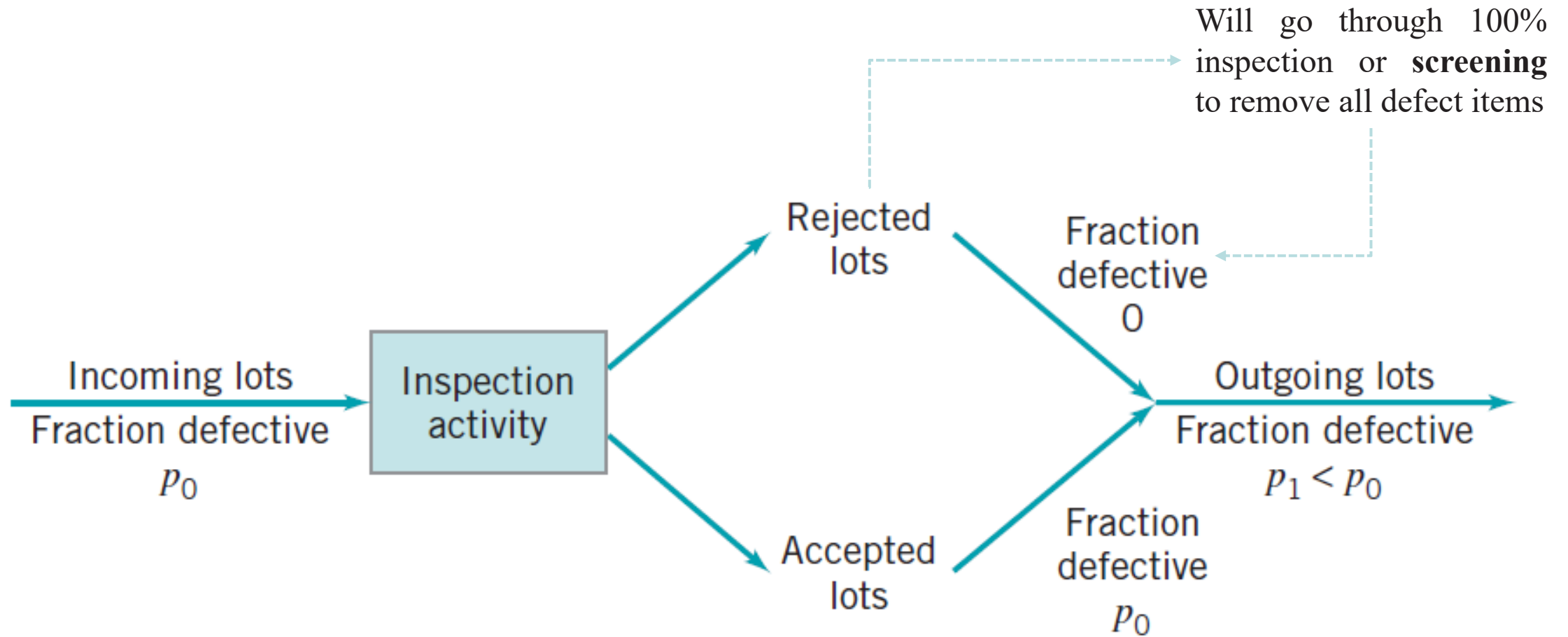
Figure 14-9 Binomial nomograph.



■ FIGURE 15.9 Binomial nomograph.

نسبة العيب كبيرة $P > 0.10$
 نسبة قبول العيب P صغير
 استعمال هذا المخطط

Rectifying Inspection Programs



A rectifying inspection program serves to “correct” lot quality.

Where to Use Rectifying Inspection

- Used when manufacturer wishes to know average level of quality that is likely to result at given stage of manufacturing
- Example stages:
 - Receiving inspection
 - In-process inspection of semi-finished goods
 - Final inspection of finished goods
- Objective: give assurance regarding average quality of material used in next stage of manufacturing operations

22-26

Average Outgoing Quality: AOQ

- Average outgoing quality (AOQ) is widely used for the evaluation of a rectifying sampling plan.
- The AOQ is the quality in the lot that results from the application of rectifying inspection.
- Average value obtained over long sequence of lots from process with fraction defective p

$$AOQ = \frac{P_a p (N - n)}{N}$$

- N - Lot size, n = number of units in sample
- Assumes all known defective units replaced with good ones, that is,
 - If lot rejected, replace all bad units in lot
 - If lot accepted, just replace the bad units in sample

Example for AOQ

To illustrate the use of equation (15.4), suppose that $N = 10,000$, $n = 89$, and $c = 2$, and that the incoming lots are of quality $p = 0.01$. Now at $p = 0.01$, we have $P_a = 0.9397$, and the AOQ is

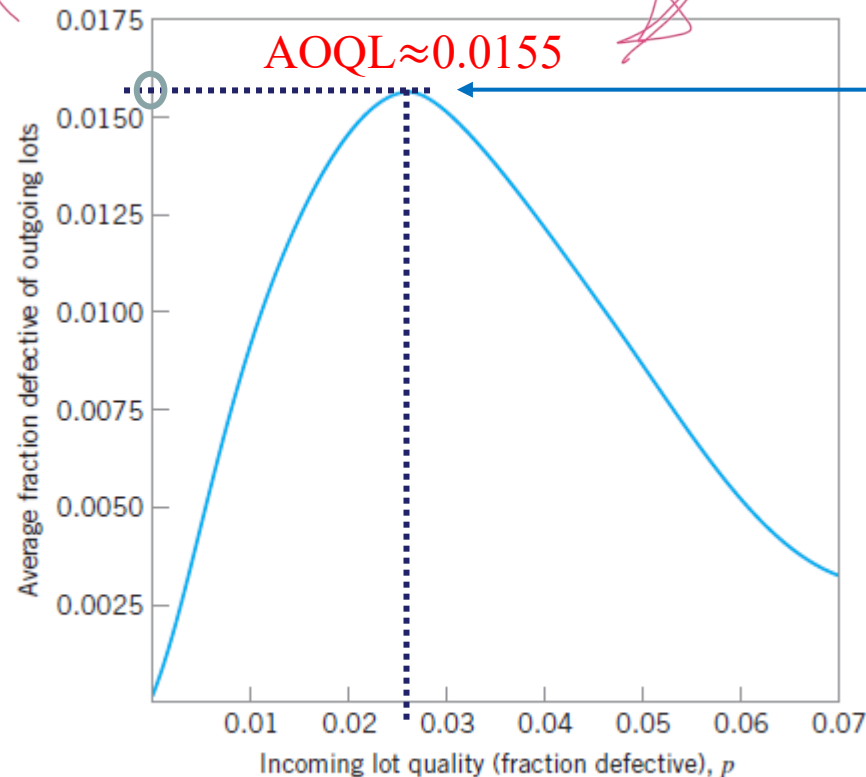
$$\begin{aligned} \text{AOQ} &= \frac{P_a p (N - n)}{N} \\ &= \frac{(0.9397)(0.01)(10,000 - 89)}{10,000} \\ &= 0.0093 \end{aligned}$$

صمكن ما تجيبها فقط تعطيني هذه
وأطلع P_a من الـ curve

That is, the average outgoing quality is 0.93% defective.

Note that as the lot size N becomes large relative to the sample size n , we may write equation 15.4 as

$$AOQ \approx P_a p \quad (15.5)$$



■ **FIGURE 15.11** Average outgoing quality curve for $n = 89, c = 2$.

The maximum ordinate on the AOQ curve represents the worst possible average quality that would result from the rectifying inspection program, and this point is called the **average outgoing quality limit (AOQL)**.

الـ Curve هو
الرسم غير مطلوب فقط اقرأه واطلع الـ Avg

AOQL is the maximum point on the curve

Another important measure relative to rectifying inspection is the total amount of inspection required by the sampling program. If the lots contain no defective items, no lots will be rejected, and the amount of inspection per lot will be the sample size n . If the items are all defective, every lot will be submitted to 100% inspection, and the amount of inspection per lot will be the lot size N . If the lot quality is $0 < p < 1$, the average amount of inspection per lot will vary between the sample size n and the lot size N . If the lot is of quality p and the probability of lot acceptance is P_a , then the **average total inspection** per lot will be

~~✗~~

$$ATI = n + (1 - P_a)(N - n)$$

(15.6)

$p_a \rightarrow$ قبول
curve ال

To illustrate the use of equation 15.6, consider our previous example with $N = 10,000$, $n = 89$, $c = 2$, and $p = 0.01$. Then, since $P_a = 0.9397$, we have

عليها مسألة

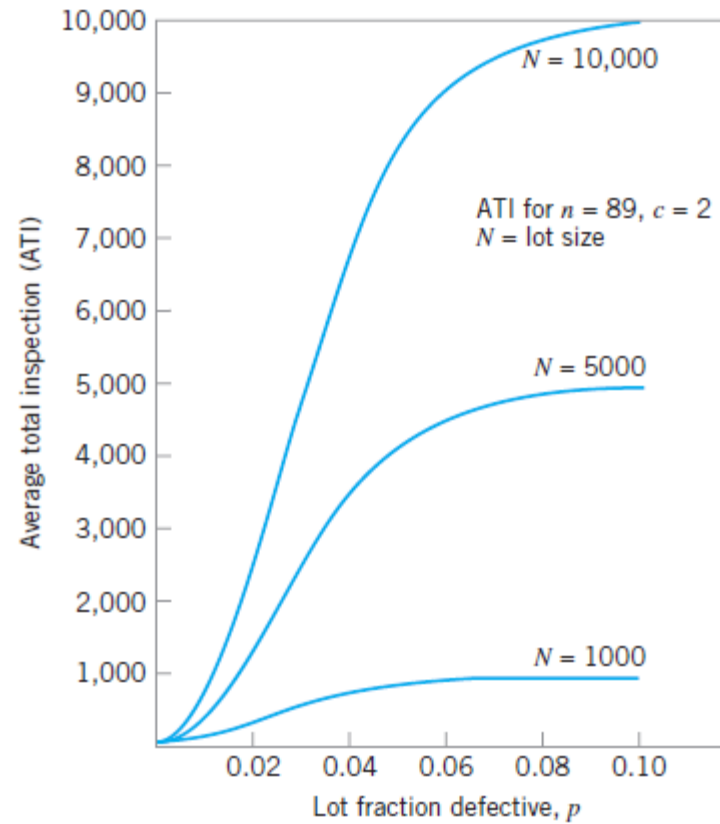
$$ATI = n + (1 - P_a)(N - n)$$

$$= 89 + (1 - 0.9397)(10,000 - 89)$$

$$= 687$$

Handwritten scribble

Curve II
also



■ **FIGURE 15.12** Average total inspection (ATI) curves for sampling plan $n = 89, c = 2$, for lot sizes of 1,000, 5,000, and 10,000.

The AOQL of a rectifying inspection plan is a very important characteristic. It is possible to design rectifying inspection programs that have specified values of AOQL. However, specification of the AOQL is not sufficient to determine a unique sampling plan. Therefore, it is relatively common practice to choose the sampling plan that has a specified AOQL and, in addition yields a minimum ATI at a particular level of lot quality. The level of lot quality usually chosen is the most likely level of incoming lot quality, which is generally called the *process average*. The procedure for generating these plans is relatively straightforward and is illustrated in Duncan (1986). Generally, it is unnecessary to go through this procedure, because tables of sampling plans that minimize ATI for a given AOQL and a specified process average p have been developed by Dodge and Romig. We describe the use of these tables in Section 15.5.

It is also possible to design a rectifying inspection program that gives a specified level of protection at the LTPD point and that minimizes the average total inspection for a specified process average p . The Dodge–Romig sampling inspection tables also provide these LTPD plans. Section 15.5 discusses the use of the Dodge–Romig tables to find plans that offer specified LTPD protection.

15.3 Double, Multiple and Sequential Sampling

A double-sampling plan is a procedure in which, under certain circumstances, a second sample is required before the lot can be sentenced. A double-sampling plan is defined by four parameters²:

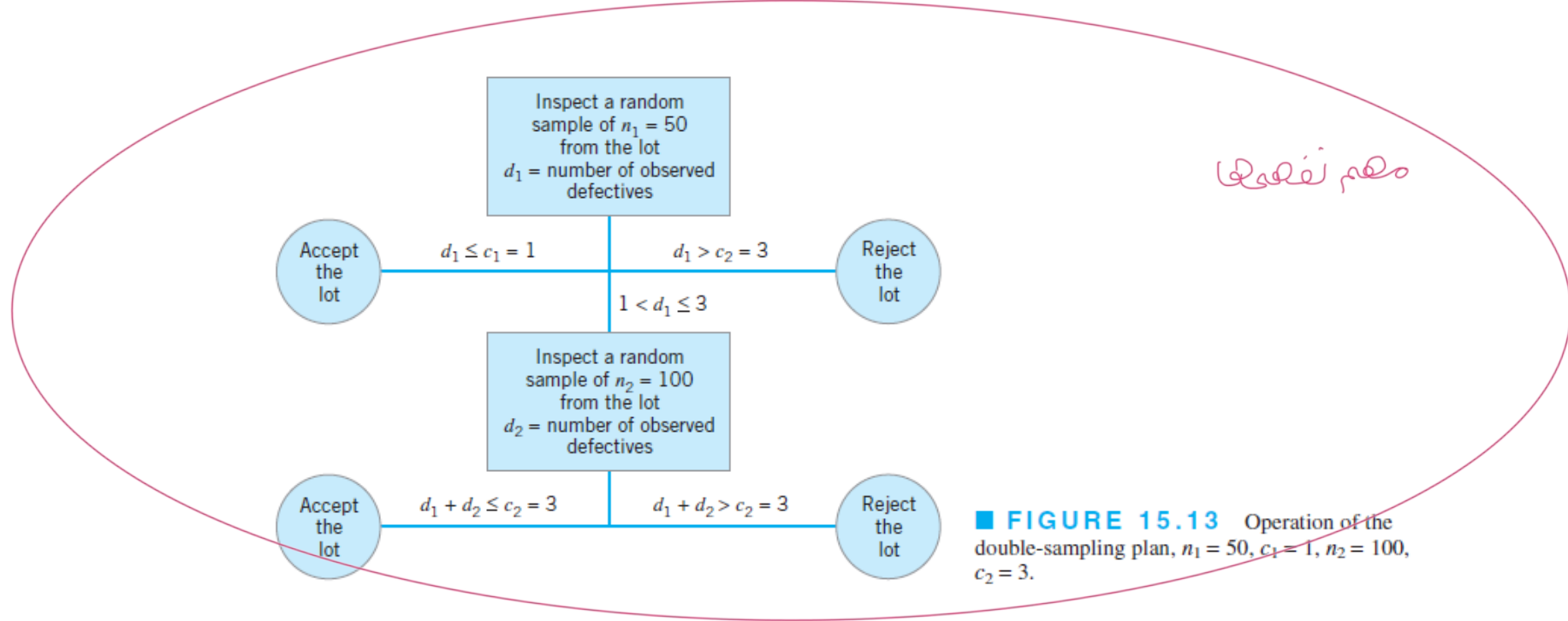
n_1 = sample size on the first sample

c_1 = acceptance number of the first sample

n_2 = sample size on the second sample

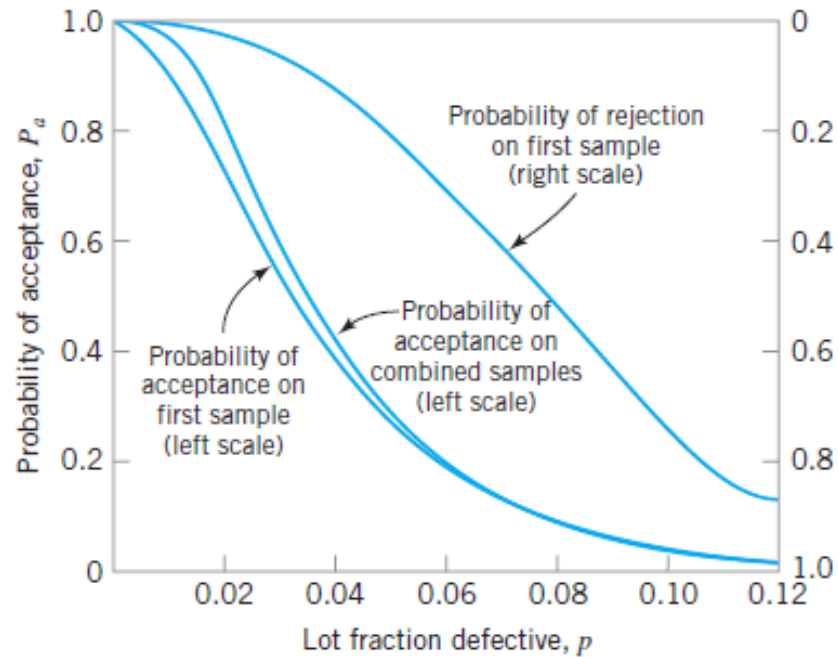
c_2 = acceptance number for both sample

As an example, suppose $n_1 = 50$, $c_1 = 1$, $n_2 = 100$, and $c_2 = 3$. Thus, a random sample of $n_1 = 50$ items is selected from the lot, and the number of defectives in the sample, d_1 , is observed. If $d_1 \leq c_1 = 1$, the lot is accepted on the first sample. If $d_1 > c_2 = 3$, the lot is rejected on the first sample. If $c_1 < d_1 \leq c_2$, a second random sample of size $n_2 = 100$ is drawn from the lot, and the number of defectives in this second sample, d_2 , is observed. Now the combined number of observed defectives from both the first and second sample, $d_1 + d_2$, is used to determine the lot sentence. If $d_1 + d_2 \leq c_2 = 3$, the lot is accepted. However, if $d_1 + d_2 > c_2 = 3$, the lot is rejected. The operation of this double-sampling plan is illustrated graphically in Figure 15.13.



- Advantage of a double-sampling plan over single sampling is that it may reduce total amount of required inspection
- Suppose first sample in a double-sampling plan is smaller than for a single-sampling plan
 - If lot is accepted or reject on first sample, cost of inspection is lower
 - Also possible to reject a lot without completing inspection of second sample

The OC Curve:



See textbook for
calculations

■ **FIGURE 15.14** OC curves for the double-sampling plan, $n_1 = 50$, $c_1 = 1$, $n_2 = 100$, $c_2 = 3$.

The Average Sample Number Curve. The average sample number curve of a double-sampling plan is also usually of interest to the quality engineer. In single sampling, the size of the sample inspected from the lot is always constant, whereas in double sampling the size of the sample selected depends on whether or not the second sample is necessary. The probability of drawing a second sample varies with the fraction defective in the incoming lot. With complete inspection of the second sample, the average sample size in double sampling is equal to the size of the first sample times the probability that there will only be one sample, plus the size of the combined samples times the probability that a second sample will be necessary. Therefore, a general formula for the average sample number in double sampling, if we assume complete inspection of the second sample, is

$$\begin{aligned} \text{ASN} &= n_1 P_1 + (n_1 + n_2)(1 - P_1) \\ &= n_1 + n_2(1 - P_1) \end{aligned} \tag{15.7}$$

where P_1 is the probability of making a lot-dispositioning decision on the *first* sample. This is

$$P_1 = P\{\text{lot is accepted on the first sample}\} + P\{\text{lot is rejected on the first sample}\}$$

If equation 15.7 is evaluated for various values of lot fraction defective p , the plot of ASN versus p is called an **average sample number curve**.

Rectifying Inspection. When rectifying inspection is performed with double sampling, the AOQ curve is given by

$$\text{AOQ} = \frac{[P_a^I(N - n_1) + P_a^{II}(N - n_1 - n_2)]p}{N} \quad (15.9)$$

assuming that all defective items discovered, either in sampling or 100% inspection, are replaced with good ones. The average total inspection curve is given by

$$\text{ATI} = n_1 P_a^I + (n_1 + n_2) P_a^{II} + N(1 - P_a) \quad (15.10)$$

Remember that $P_a = P_a^I + P_a^{II}$ is the probability of final lot acceptance and that the acceptance probabilities depend on the level of lot or process quality p .

15.3.2 Multiple-Sampling Plans

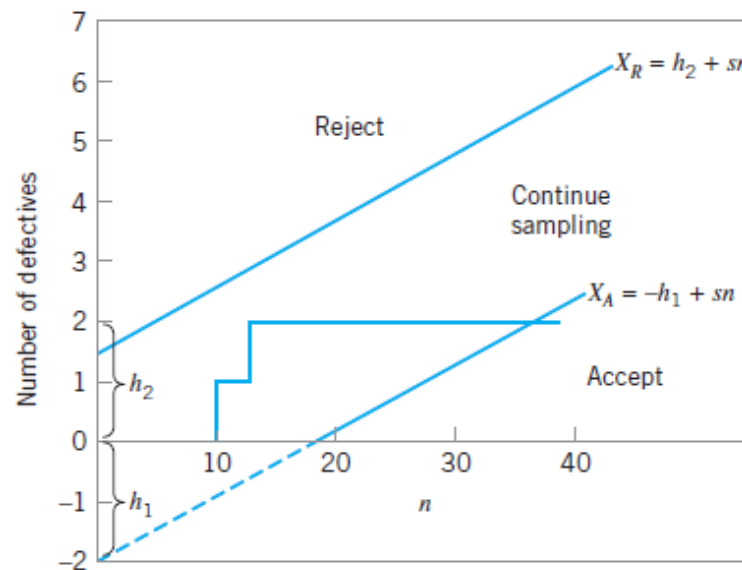
A multiple-sampling plan is an extension of double sampling in that more than two samples can be required to sentence a lot. An example of a multiple-sampling plan with five stages follows.

Cumulative Sample Size	Acceptance Number	Rejection Number
20	0	3
40	1	4
60	3	5
80	5	7
100	8	9

This plan will operate as follows: If, at the completion of any stage of sampling, the number of defective items is less than or equal to the acceptance number, the lot is accepted. If, during any stage, the number of defective items equals or exceeds the rejection number, the lot is rejected; otherwise the next sample is taken. The multiple-sampling procedure continues until the fifth sample is taken, at which time a lot disposition decision must be made. The first sample is usually inspected 100%, although subsequent samples are usually subject to curtailment.

15.3.3 Sequential-Sampling Plans

Sequential sampling is an extension of the double-sampling and multiple-sampling concept. In sequential sampling, we take a sequence of samples from the lot and allow the number of samples to be determined entirely by the results of the sampling process. In practice, sequential sampling can theoretically continue indefinitely, until the lot is inspected 100%. In practice, sequential sampling plans are usually truncated after the number inspected is equal to three times the number that would have been inspected using a corresponding single-sampling plan. If the sample size selected at each stage is greater than one, the process is usually called *group* sequential sampling. If the sample size inspected at each stage is 1, the procedure is usually called **item-by-item sequential sampling**.



■ **FIGURE 15.16** Graphical performance of sequential sampling.

$$X_A = -h_1 + sn \quad (\text{acceptance line}) \quad (15.11a)$$

$$X_R = h_2 + sn \quad (\text{rejection line}) \quad (15.11b)$$

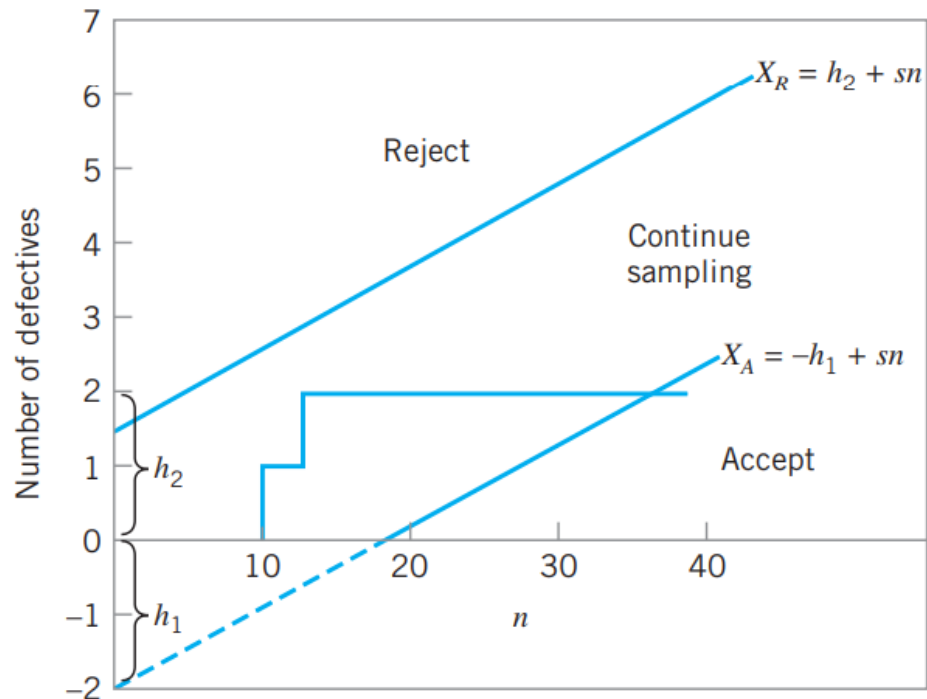
where

$$h_1 = \left(\log \frac{1-\alpha}{\beta} \right) / k \quad (15.12)$$

$$h_2 = \left(\log \frac{1-\beta}{\alpha} \right) / k \quad (15.13)$$

$$k = \log \frac{p_2(1-p_1)}{p_1(1-p_2)} \quad (15.14)$$

$$s = \log \left[(1-p_1)/(1-p_2) \right] / k \quad (15.15)$$



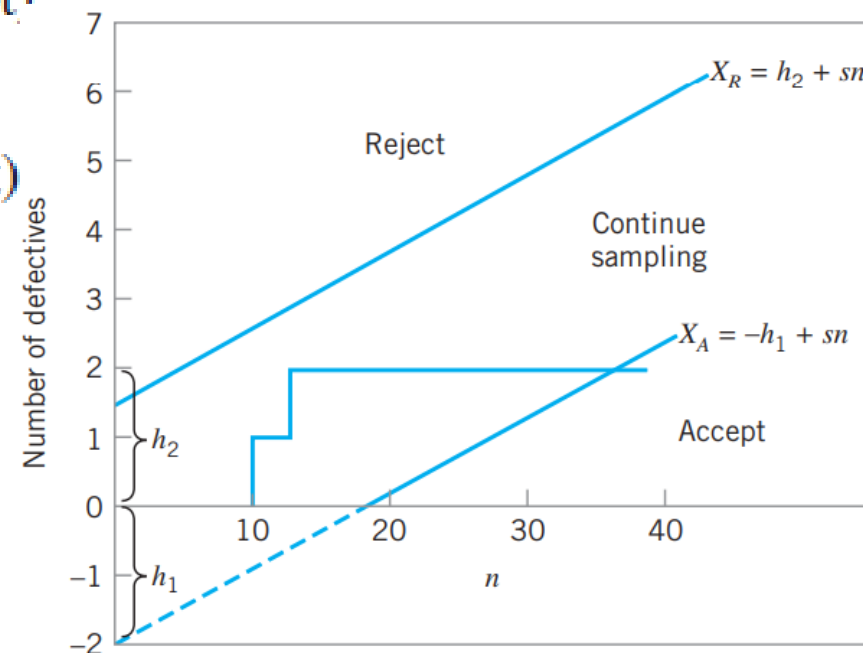
To illustrate the use of these equations, suppose we wish to find a sequential-sampling plan for which $p_1 = 0.01$, $\alpha = 0.05$, $p_2 = 0.06$, and $\beta = 0.10$. Thus,

$$\begin{aligned}k &= \log \frac{p_2(1-p_1)}{p_1(1-p_2)} \\&= \log \frac{(0.06)(0.99)}{(0.01)(0.94)} = 0.80066 \\h_1 &= \left(\log \frac{1-\alpha}{\beta} \right) / k \\&= \left(\log \frac{0.95}{0.10} \right) / 0.80066 = 1.22 \\h_2 &= \left(\log \frac{1-\beta}{\alpha} \right) / k \\&= \left(\log \frac{0.90}{0.05} \right) / 0.80066 = 1.57 \\s &= \log [(1-p_1)/(1-p_2)] / k \\&= [\log(0.99/0.94)] / 0.80066 = 0.028\end{aligned}$$

Instead of using a graph to determine the lot disposition, the sequential-sampling plan can be displayed in a table such as Table 15.3. The entries in the table are found by substituting values of n into the equations for the acceptance and rejection lines and calculating acceptance and rejection numbers. For example, the calculations for $n = 45$ are

$$\begin{aligned} X_A &= -1.22 + 0.028n \\ &= -1.22 + 0.028(45) = 0.04 \quad (\text{accept}) \end{aligned}$$

$$\begin{aligned} X_R &= 1.57 + 0.028n \\ &= 1.57 + 0.028(45) = 2.83 \quad (\text{reject}) \end{aligned}$$



The OC Curve and ASN Curve for Sequential Sampling. The OC curve for sequential sampling can be easily obtained. Two points on the curve are $(p_1, 1 - \alpha)$ and (p_2, β) . A third point, near the middle of the curve, is $p = s$ and $P_a = h_2/(h_1 + h_2)$.

The average sample number taken under sequential-sampling is

$$\text{ASN} = P_a \left(\frac{A}{C} \right) + (1 - P_a) \frac{B}{C} \quad (15.16)$$

where

$$A = \log \frac{\beta}{1 - \alpha}$$

$$B = \log \frac{1 - \beta}{\alpha}$$

and

$$C = p \log \left(\frac{p_2}{p_1} \right) + (1 - p) \log \left(\frac{1 - p_2}{1 - p_1} \right)$$

Rectifying Inspection. The average outgoing quality (AOQ) for sequential-sampling is given approximately by

$$\text{AOQ} \simeq P_a p \quad (15.17)$$

The average total inspection is also easily obtained. Note that the amount of sampling is A/C when a lot is accepted and N when it is rejected. Therefore, the average total inspection is

$$\text{ATI} = P_a \left(\frac{A}{C} \right) + (1 - P_a) N \quad (15.18)$$

15.4 Military Standard 105E (ANSI/ASQC Z1.4, ISO 2859)

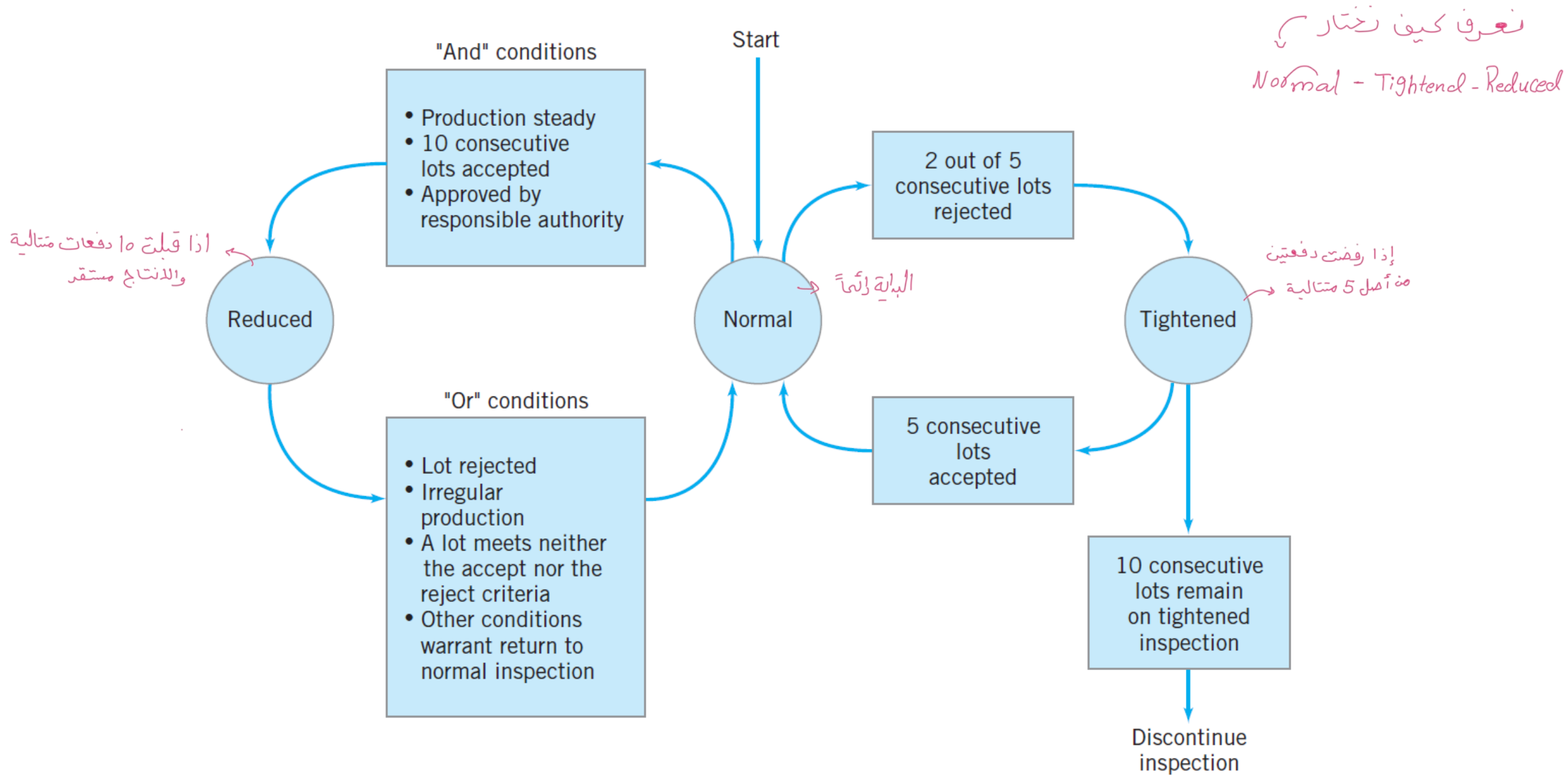
15.4.1 Description of the Standard

- Standard sampling procedures for inspection by attributes were developed during World War II.
- MIL STD 105E is the most widely used acceptance-sampling system for attributes in the world today.
- Provides for three types of sampling: single, double, and multiple
- Primary focal point is the acceptable quality level (AQL)
 - Different AQLs may be designated for different types of defects: critical, major, and minor
- Sample size is determined by **lot size** and by choice of **inspection level**
- There is a derivative civilian standard, ANSI/ASQC Z1.4, which is quite similar to the military standard. The standard was also adopted by the International Organization for Standardization as ISO 2859.

15.4.1

- In sampling plan, a provision is made for either **normal inspection, tightened inspection, or reduced inspection**.
 - **Normal inspection** is used at the start of the inspection activity.
 - **Tightened inspection** is instituted when the supplier's recent quality history has deteriorated. Acceptance requirements for lots under tightened inspection are more stringent than under normal inspection.
 - **Reduced inspection** is instituted when the supplier's recent quality history has been exceptionally good. The sample size generally used under reduced inspection is less than that under normal inspection.

- The sample size used in MIL STD 105E is determined by the lot size and by the choice of inspection level.
- Three general levels of inspection are provided.
 - Level II is designated as normal (Widely used).
 - Level I requires about one-half the amount of inspection as Level II and may be used when less discrimination is needed.
 - Level III requires about twice as much inspection as Level II and should be used when more discrimination is needed.
- There are also four special inspection levels: S-1, S-2, S-3, and S-4. The special inspection levels use very small samples, and should be employed only when the small sample sizes are necessary and when greater sampling risks can or must be tolerated.



■ **FIGURE 15.17** Switching rules for normal, tightened, and reduced inspection, MIL STD 105E.

15.4.2 Procedure



صفحة والجدول معطى

A step-by-step procedure for using MIL STD 105E is as follows:

1. Choose the AQL.
2. Choose the inspection level.
3. Determine the lot size.
4. Find the appropriate sample size code letter from Table 15.4.
5. Determine the appropriate type of sampling plan to use (single, double, multiple).
6. Enter the appropriate table to find the type of plan to be used.
7. Determine the corresponding normal and reduced inspection plans to be used when required.

Table 15.4 presents the sample size code letters for MIL STD 105E. Tables 15.5, 15.6, and 15.7 present the single-sampling plans for normal inspection, tightened inspection, and reduced inspection, respectively.

■ **TABLE 15.4**

Sample Size Code Letters (MIL STD 105E, Table 1)

Lot or Batch Size	Special Inspection Levels				General Inspection Levels		
	S-1	S-2	S-3	S-4	I	II	III
2 to 8	A	A	A	A	A	A	B
9 to 15	A	A	A	A	A	B	C
16 to 25	A	A	B	B	B	C	D
26 to 50	A	B	B	C	C	D	E
51 to 90	B	B	C	C	C	E	F
91 to 150	B	B	C	D	D	F	G
151 to 280	B	C	D	E	E	G	H
281 to 500	B	C	D	E	F	H	J
501 to 1,200	C	C	E	F	G	J	K
1,201 to 3,200	C	D	E	G	H	K	L
3,201 to 10,000	C	D	F	G	J	L	M
10,001 to 35,000	C	D	F	H	K	M	N
35,001 to 150,000	D	E	G	J	L	N	P
150,001 to 500,000	D	E	G	J	M	P	Q
500,001 and over	D	E	H	K	N	Q	R

■ **TABLE 15.5**

Master Table for Normal Inspection for Single Sampling (U.S. Dept. of Defense MIL STD 105E, Table II-A)

Sample Size Code Letter	Sample Size	Acceptable Quality Levels (normal inspection)																																																																																																																																																																																																																																										
		0.010	0.015	0.025	0.040	0.065	0.10	0.15	0.25	0.40	0.65	1.0	1.5	2.5	4.0	6.5	10	15	25	40	65	100	150	250	400	650	1000																																																																																																																																																																																																																	
		Ac Re	Ac Re	Ac Re	Ac Re	Ac Re	Ac Re	Ac Re	Ac Re	Ac Re	Ac Re	Ac Re	Ac Re	Ac Re	Ac Re	Ac Re	Ac Re	Ac Re	Ac Re	Ac Re	Ac Re	Ac Re	Ac Re	Ac Re	Ac Re	Ac Re	Ac Re	Ac Re																																																																																																																																																																																																																
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P	800																																																																																																																																																																																																																																											
Q	1250																																																																																																																																																																																																																																											
R	2000																																																																																																																																																																																																																																											

↓ = Use first sampling plan below arrow. If sample size equals, or exceeds, lot or batch size, do 100% inspection.

↑ = Use first sampling plan above arrow.

Ac = Acceptance number.

Re = Rejection number.

■ **TABLE 15.6**

Master Table for Tightened Inspection for Single Sampling (U.S. Dept. of Defense MIL STD 105E, Table II-B)

Sample Size Code Letter	Sample Size	Acceptable Quality Levels (tightened inspection)																											
		0.010	0.015	0.025	0.040	0.065	0.10	0.15	0.25	0.40	0.65	1.0	1.5	2.5	4.0	6.5	10	15	25	40	65	100	150	250	400	650	1000		
		Ac Re	Ac Re	Ac Re	Ac Re	Ac Re	Ac Re	Ac Re	Ac Re	Ac Re	Ac Re	Ac Re	Ac Re	Ac Re	Ac Re	Ac Re	Ac Re	Ac Re	Ac Re	Ac Re	Ac Re	Ac Re	Ac Re	Ac Re	Ac Re	Ac Re	Ac Re	Ac Re	Ac Re
A	2	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	1 2	2 3	3 4	5 6	8 9	12 13	18 19	27 28		
B	3	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	0 1	↓	↓	1 2	2 3	3 4	5 6	8 9	12 13	18 19	27 28	41 42		
C	5	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	0 1	↓	↓	1 2	2 3	3 4	5 6	8 9	12 13	18 19	27 28	41 42	↑		
D	8	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	0 1	↓	↓	1 2	2 3	3 4	5 6	8 9	12 13	18 19	27 28	41 42	↑	↑		
E	13	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	0 1	↓	↓	1 2	2 3	3 4	5 6	8 9	12 13	18 19	27 28	41 42	↑	↑	↑		
F	20	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	0 1	↓	↓	1 2	2 3	3 4	5 6	8 9	12 13	18 19	↑	↑	↑	↑	↑	↑		
G	32	↓	↓	↓	↓	↓	↓	↓	↓	↓	0 1	↓	↓	1 2	2 3	3 4	5 6	8 9	12 13	18 19	↑	↑	↑	↑	↑	↑	↑		
H	50	↓	↓	↓	↓	↓	↓	↓	0 1	↓	↓	1 2	2 3	3 4	5 6	8 9	12 13	18 19	↑	↑	↑	↑	↑	↑	↑	↑	↑		
J	80	↓	↓	↓	↓	↓	↓	0 1	↓	↓	1 2	2 3	3 4	5 6	8 9	12 13	18 19	↑	↑	↑	↑	↑	↑	↑	↑	↑	↑		
K	125	↓	↓	↓	↓	↓	0 1	↓	↓	1 2	2 3	3 4	5 6	8 9	12 13	18 19	↑	↑	↑	↑	↑	↑	↑	↑	↑	↑	↑		
L	200	↓	↓	↓	↓	0 1	↓	↓	1 2	2 3	3 4	5 6	8 9	12 13	18 19	↑	↑	↑	↑	↑	↑	↑	↑	↑	↑	↑	↑		
M	315	↓	↓	↓	↓	0 1	↓	↓	1 2	2 3	3 4	5 6	8 9	12 13	18 19	↑	↑	↑	↑	↑	↑	↑	↑	↑	↑	↑	↑		
N	500	↓	↓	0 1	↓	↓	1 2	2 3	3 4	5 6	8 9	12 13	18 19	↑	↑	↑	↑	↑	↑	↑	↑	↑	↑	↑	↑	↑	↑		
P	800	↓	0 1	↓	↓	1 2	2 3	3 4	5 6	8 9	12 13	18 19	↑	↑	↑	↑	↑	↑	↑	↑	↑	↑	↑	↑	↑	↑	↑		
Q	1250	↓	0 1	↓	↓	1 2	2 3	3 4	5 6	8 9	12 13	18 19	↑	↑	↑	↑	↑	↑	↑	↑	↑	↑	↑	↑	↑	↑	↑		
R	2000	0 1	↑	↓	1 2	2 3	3 4	5 6	8 9	12 13	18 19	↑	↑	↑	↑	↑	↑	↑	↑	↑	↑	↑	↑	↑	↑	↑	↑		
S	3150			1 2																									

↓ = Use first sampling plan below arrow. If sample size equals, or exceeds, lot or batch size, do 100% inspection.

↑ = Use first sampling plan above arrow.

Ac = Acceptance number.

Re = Rejection number.

■ TABLE 15.7

Master Table for Reduced Inspection for Single Sampling (U.S. Dept. of Defense MIL STD 105E, Table II-C)

Sample Size Code Letter	Sample Size	Acceptable Quality Levels (reduced inspection)†																											
		0.010	0.015	0.025	0.040	0.065	0.10	0.15	0.25	0.40	0.65	1.0	1.5	2.5	4.0	6.5	10	15	25	40	65	100	150	250	400	650	1000		
		Ac Re	Ac Re	Ac Re	Ac Re	Ac Re	Ac Re	Ac Re	Ac Re	Ac Re	Ac Re	Ac Re	Ac Re	Ac Re	Ac Re	Ac Re	Ac Re	Ac Re	Ac Re	Ac Re	Ac Re	Ac Re	Ac Re	Ac Re	Ac Re	Ac Re	Ac Re	Ac Re	
A	2	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓		
B	2	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓		
C	2	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓		
D	3	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓		
E	5	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓		
F	8	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓		
G	13	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓		
H	20	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓		
J	32	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓		
K	50	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓		
L	80	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓		
M	125	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓		
N	200	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓		
P	315	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓		
Q	500	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓		
R	800	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓		

↓ = Use first sampling plan below arrow. If sample size equals, or exceeds, lot or batch size, do 100% inspection.

↑ = Use first sampling plan above arrow.

Ac = Acceptance number.

Re = Rejection number.

† = If the acceptance number has been exceeded, but the rejection number has not been reached, accept the lot, but reinstate normal inspection.

Example



To illustrate the use of MIL STD 105E, suppose that a product is submitted in lots of size $N = 2,000$. The acceptable quality level is 0.65%.

- Use the standard to generate normal, single-sampling plans for this situation.

A step-by-step procedure for using MIL STD 105E is as follows:

1. Choose the AQL. $AQL = 0.0065$ or 0.65% .
2. Choose the inspection level. Normal inspection
3. Determine the lot size. $N = 2,000$
4. Find the appropriate sample size code letter from Table 15.4.
5. Determine the appropriate type of sampling plan to use (single, double, multiple).
6. Enter the appropriate table to find the type of plan to be used.
7. Determine the corresponding normal and reduced inspection plans to be used when required.

Table 15.4 presents the sample size code letters for MIL STD 105E. Tables 15.5, 15.6, and 15.7 present the single-sampling plans for normal inspection, tightened inspection, and reduced inspection, respectively.

Example

■ **TABLE 15.4**

Sample Size Code Letters (MIL STD 105E, Table 1)

Lot or Batch Size	Special Inspection Levels				General Inspection Levels		
	S-1	S-2	S-3	S-4	I	II	III
2 to 8	A	A	A	A	A	A	B
9 to 15	A	A	A	A	A	B	C
16 to 25	A	A	B	B	B	C	D
26 to 50	A	B	B	C	C	D	E
51 to 90	B	B	C	C	C	E	F
91 to 150	B	B	C	D	D	F	G
151 to 280	B	C	D	E	E	G	H
281 to 500	B	C	D	E	F	H	J
501 to 1,200	C	C	E	F	G	J	K
1,201 to 3,200	C	D	E	G	H	K	L
3,201 to 10,000	C	D	F	G	J	L	M
10,001 to 35,000	C	D	F	H	K	M	N
35,001 to 150,000	D	E	G	J	L	N	P
150,001 to 500,000	D	E	G	J	M	P	Q
500,001 and over	D	E	H	K	N	Q	R

$N = 2,000$

The sample size code letter = K

Refer to Tables 15.5, -6 and -7 in the textbook on pp. 658 – 660.

Example

A step-by-step procedure for using MIL STD 105E is as follows:

1. Choose the AQL. $AQL = 0.0065$ or 0.65% .
2. Choose the inspection level. Normal inspection
3. Determine the lot size. $N = 2,000$
4. Find the appropriate sample size code letter from Table 15.4. The sample size code letter is K
5. Determine the appropriate type of sampling plan to use (single, double, multiple). single sampling
6. Enter the appropriate table to find the type of plan to be used.
7. Determine the corresponding normal and reduced inspection plans to be used when required.

Table 15.4 presents the sample size code letters for MIL STD 105E. Tables 15.5, 15.6, and 15.7 present the single-sampling plans for normal inspection, tightened inspection, and reduced inspection, respectively.

Example

TABLE 15.5

Master Table for Normal Inspection for Single Sampling (U.S. Dept. of Defense MIL STD 105E, Table II-A)

Sample Size Code Letter	Sample Size	Acceptable Quality Levels (normal inspection)																											
		0.010	0.015	0.025	0.040	0.065	0.10	0.15	0.25	0.40	0.65	1.0	1.5	2.5	4.0	6.5	10	15	25	40	65	100	150	250	400	650	1000		
		Ac Re	Ac Re	Ac Re	Ac Re	Ac Re	Ac Re	Ac Re	Ac Re	Ac Re	Ac Re	Ac Re	Ac Re	Ac Re	Ac Re	Ac Re	Ac Re	Ac Re	Ac Re	Ac Re	Ac Re	Ac Re	Ac Re	Ac Re	Ac Re	Ac Re	Ac Re	Ac Re	
A	2	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	0 1	↓	↓	1 2	2 3	3 4	5 6	7 8	10 11	14 15	21 22	30 31	44 45		
B	3	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	0 1	↑	↓	1 2	2 3	3 4	5 6	7 8	10 11	14 15	21 22	30 31				
C	5	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	0 1	↑	↓	1 2	2 3	3 4	5 6	7 8	10 11	14 15	21 22	30 31				
D	8	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	0 1	↑	↓	1 2	2 3	3 4	5 6	7 8	10 11	14 15	21 22	30 31	44 45			
E	13	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↑	↓	1 2	2 3	3 4	5 6	7 8	10 11	14 15	21 22	30 31	44 45	↑	↑		
F	20	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↑	↓	1 2	2 3	3 4	5 6	7 8	10 11	14 15	21 22	↑	↑				
G	32	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	1 2	2 3	3 4	5 6	7 8	10 11	14 15	21 22	↑	↑	↑	↑			
H	50	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	2 3	3 4	5 6	7 8	10 11	14 15	21 22	↑	↑	↑					
J	80	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	3 4	5 6	7 8	10 11	14 15	21 22	↑	↑	↑	↑					
K	125	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	4 5	6 7	8 10	11 14	15 21	22 30	↑	↑	↑	↑	↑	↑	↑		
L	200	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	5 6	7 8	10 11	14 15	21 22	↑	↑	↑	↑	↑	↑	↑	↑		
M	315	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	6 7	8 10	11 14	15 21	22 30	↑	↑	↑	↑	↑	↑				
N	500	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	7 8	10 11	14 15	21 22	↑	↑	↑	↑	↑	↑	↑				
P	800	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	8 10	11 14	15 21	22 30	↑	↑	↑	↑	↑	↑	↑				
Q	1250	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	9 10	11 14	15 21	22 30	↑	↑	↑	↑	↑	↑	↑	↑			
R	2000	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	10 11	14 15	21 22	↑	↑	↑	↑	↑	↑	↑					

↓ = Use first sampling plan below arrow. If sample size equals, or exceeds, lot or batch size, do 100% inspection.

↑ = Use first sampling plan above arrow.

Ac = Acceptance number.

Re = Rejection number.

$n = 125$

Acceptance number $c=2$

Rejection number $r=3$

Problem 15.22

A product is supplied in lots of size $N = 10,000$. The AQL has been specified at 0.10%.

- Find the normal single-sampling plans from MIL STD 105E, assuming general inspection level II.

$N = 10,000$; AQL = 0.10%; General inspection level II; Sample size code letter = L

■ TABLE 15.5

Master Table for Normal Inspection for Single Sampling (U.S. Dept. of Defense MIL STD 105E, Table II-A)

Sample Size Code Letter	Sample Size	Acceptable Quality Levels (normal inspection)																											
		0.010	0.015	0.025	0.040	0.065	0.10	0.15	0.25	0.40	0.65	1.0	1.5	2.5	4.0	6.5	10	15	25	40	65	100	150	250	400	650	1000		
		Ac Re	Ac Re	Ac Re	Ac Re	Ac Re	Ac Re	Ac Re	Ac Re	Ac Re	Ac Re	Ac Re	Ac Re	Ac Re	Ac Re	Ac Re	Ac Re	Ac Re	Ac Re	Ac Re	Ac Re	Ac Re	Ac Re	Ac Re	Ac Re	Ac Re	Ac Re	Ac Re	
A	2														0 1					1 2	2 3	3 4	5 6	7 8	10 11	14 15	21 22	30 31	
B	3														0 1				1 2	2 3	3 4	5 6	7 8	10 11	14 15	21 22	30 31	44 45	
C	5																1 2	2 3	3 4	5 6	7 8	10 11	14 15	21 22	30 31	44 45			
D	8													0 1				2 3	3 4	5 6	7 8	10 11	14 15	21 22	30 31	44 45			
E	13										0 1						1 2	2 3	3 4	5 6	7 8	10 11	14 15	21 22	30 31	44 45			
F	20									0 1						1 2	2 3	3 4	5 6	7 8	10 11	14 15	21 22						
G	32								0 1					1 2	2 3	3 4	5 6	7 8	10 11	14 15	21 22								
H	50							0 1					1 2	2 3	3 4	5 6	7 8	10 11	14 15	21 22									
J	80										1 2			2 3	3 4	5 6	7 8	10 11	14 15	21 22									
K	125						0 1			1 2	2 3	3 4	5 6	7 8	10 11	14 15	21 22												
L	200					0 1			1 2	2 3	3 4	5 6	7 8	10 11	14 15	21 22													
M	315				0 1			1 2	2 3	3 4	5 6	7 8	10 11	14 15	21 22														
N	500			0 1			1 2	2 3	3 4	5 6	7 8	10 11	14 15	21 22															
P	800		0 1			1 2	2 3	3 4	5 6	7 8	10 11	14 15	21 22																
Q	1250	0 1				2 3	3 4	5 6	7 8	10 11	14 15	21 22																	
R	2000			1 2	2 3	3 4	5 6	7 8	10 11	14 15	21 22																		

$n = 125$

Acceptance number $c=0$

Rejection number $r=1$

↓ = Use first sampling plan below arrow. If sample size equals, or exceeds, lot or batch size, do 100% inspection.

↑ = Use first sampling plan above arrow.

Ac = Acceptance number.

Re = Rejection number.

15.4.3 Discussion

- Several points about MIL STD 105E should be emphasized:
- MIL STD 105E is AQL-oriented
- Not all possible sample sizes are possible (2,3,5,8,13,20,32,50, etc.)
- Sample sizes are related to lot sizes
- Switching rules are subject to criticism for both miss witching between inspection plans and discontinuation even though there has been no actual quality deterioration
- But a **flagrant** and **common** abuse of MIL STD 105E is failure to use the switching rules at all